Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of an Integrated Multi-Functional Nylon/Cotton Materials-to-Garment System for Next-Generation Military Uniforms**			
Korea's 12 major national strategic technology fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification 1)	Technology Field 1 (Major Category)		Technology Field 2 (Major Category)	
	Chemicals			
	Medium category	Small category	Medium category	Small category
	Fiber Manufacturing	*Fiber Spinning*		
Proposed technology concept	**Core Concept** The proposed technology aims to develop an integrated materials-to-garment system for military uniforms that provides flame retardancy, infrared (IR) camouflage, durability, and adaptive thermal management across functional NC (Nylon/Cotton) combat uniforms and environment-adaptive thermal innerwear. The core concept is to establish a full-cycle materials and textile engineering system from fiber/blended yarn design, functional fabric development, hybrid finishing, garment prototyping to field-oriented validation, that is capable of meeting next-generation military operational requirements. **Key Functions and Potential Applications** **Key Functions**			

- Intrinsic Flame-Retardant Performance
  - Aramid/Cotton/Nylon blended yarns with embedded flame-retardant properties
  - Reduced dependency on finishing-based FR treatment, ensuring long-term stability
  - Formation of protective carbonized layers under heat exposure to suppress flame spread
  - FR performance maintained after repeated washing, abrasion, and field use
- Infrared (IR) Camouflage Capability
  - Functional dyes/pigments and controlled-reflectance coatings optimized for 700–1200 nm
  - Stable IR signature under various environmental and mission conditions
  - Long-term retention of IR performance through inorganic–organic hybrid finishing
- Adaptive Thermal Heating and Insulation
  - Hygroscopic heating, photothermal absorption, structural insulation, and optional active heating
  - Configurable thermal response for cold-weather operations and long-duration missions
  - Heating performance preserved after multiple washing cycles and extended wear
- Extreme-Environment Response for Special Uniforms
  - Fabric structures engineered for high-temperature, enclosed, and long-duration operational settings
  - Heat/moisture management to mitigate thermal accumulation
  - Balanced mechanical strength, flexibility, and weight for confined environments
- Durability and Operational Reliability
  - Functional stability under >10 washing cycles, abrasion, heat exposure, and long-term wear
  - Simultaneous retention of FR, IR, thermal, and mechanical performance in field conditions

Potential Applications

- Military Applications
  - NC combat uniforms (summer and winter variants)
  - Combat shirts requiring breathability, FR, and IR camouflage
  - Tracked-vehicle crew uniforms requiring heat, humidity, and enclosed-environment optimization
  - Environment-adaptive thermal innerwear for cold-weather, high-altitude, or prolonged missions
- Defense and National Security Extensions
  - Special operations apparel with integrated FR/IR/thermal requirements
  - Protective garments for vehicle crews, artillery units, logistics personnel, and maintenance crews

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	● Civilian / Dual-Use Applications ■ Disaster-response and firefighter secondary protective gear ■ Industrial safety clothing for high-heat or hazardous environments ■ Arctic/Antarctic or extreme-climate workwear ■ Functional outdoor and mountaineering apparel
Rationale for Joint R&D Project	**Critical Technical Barriers** ● High Complexity of Multi-Functional Integration Achieving simultaneous flame retardancy, IR camouflage, thermal responsiveness, durability, and comfort presents inherent trade-offs. ■ FR additives can interfere with IR signatures and dye uptake. ■ IR-reflectance control may conflict with breathability or thermal insulation. ■ Thermal heating structures may compromise abrasion resistance or flexibility. ■ Developing a harmonized system that balances these competing requirements remains a major technical barrier. ● Challenges in Blended-Yarn Intrinsic FR Design Intrinsic flame-retardant performance requires precise control of Aramid/Cotton/Nylon blended yarn composition, fiber dispersion, and spinning conditions. ■ Incompatibility between fiber types complicates spinnability and yarn uniformity. ■ Maintaining FR stability while preserving textile hand, flexibility, and strength is difficult. ■ Reducing reliance on FR finishing demands breakthroughs at the fiber and yarn design level. ● Infrared Camouflage Material Stability and Performance Control Military-grade IR camouflage requires accurate reflectance control in the 700–1200 nm wavelength range. ■ Developing stable IR dyes/pigments with low degradation under UV, washing, sweat, and abrasion is challenging. ■ Ensuring uniform IR performance across multi-layer garments and production batches is difficult. ■ Integrating IR materials without compromising mechanical or FR performance poses a significant barrier. ● Durability of Hybrid Organic–Inorganic Finishing Hybrid finishing used to enhance FR, IR, and thermal characteristics introduces additional durability challenges. ■ Maintaining coating adhesion and flexibility on NC-based fabrics is difficult. ■ Preventing cracking, delamination, and performance loss after repeated washing or abrasion is essential. ■ Achieving scalable, reproducible hybrid-finishing processes that retain long-term functionality is technically demanding. ● Thermal and Moisture Management for Extreme Environments For applications such as tracked-vehicle crew uniforms, fabrics must simultaneously.

- Withstand elevated temperatures and heat buildup
- Prevent sweat accumulation and thermal entrapment in confined spaces
- Maintain structural integrity under prolonged and enclosed-operation wear

  Designing such multi-layer, multi-functional textile structures is technically challenging.

- Long-Term Durability Under Field-Relevant Stressors

  Military apparel must retain performance after:

  - Repeated washing (>10 cycles)
  - Sweat, abrasion, UV exposure, and heat loads
  - Long-duration wear under load and garment deformation

    Ensuring long-term preservation of FR, IR, and heating properties in realistic operational conditions remains a significant barrier.

- Integrated Field-Oriented Verification and Testing Limitations

  A major challenge is the lack of a unified testing protocol capable of evaluating multi-functional performance under real operational conditions.

  - Simulating combined washing–abrasion–heat–humidity stress modes
  - Establishing standardized test metrics for IR signature stability
  - Generating long-term durability data for FR and thermal functions
  - Developing such a validation framework is critical but technically difficult.

- Scalability and Production Reproducibility

  To transition to military procurement, the technology must satisfy strict quality uniformity requirements.

  - Scaling blended yarn production while maintaining consistent FR and IR properties
  - Variation in dyeing/finishing outcomes between production batches
  - Ensuring reproducibility across different manufacturers and supply-chain environments

Why Global Collaboration?

Developing advanced military garments that combine flame retardancy, IR camouflage, durability, and adaptive thermal performance requires materials research and validation capabilities that are not fully available within typical domestic manufacturing environments. While Korean textile SMEs are highly competitive in production, processing, and rapid product development, achieving intrinsic functional performance, particularly in blended-yarn FR design, IR spectral control, and hybrid finishing stability, that benefits from more specialized materials science expertise.

Purdue University has complementary strengths in fiber and composite materials, IR-related textile characterization, and thermal–moisture behavior analysis. By combining Korean industry's manufacturing and commercialization capabilities with Purdue's research and validation expertise, the project can establish a more reliable technical foundation and develop functional garments that meet higher performance standards required for military applications. This collaboration ensures that the resulting technologies are both scientifically validated and practically scalable.

Key words	1. Flame-Retardant Military Textiles

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	2. Infrared (IR) Camouflage Fabrics 3. Blended-Yarn Functional Materials 4. Hybrid Organic–Inorganic Finishing 5. Adaptive Thermal Innerwear

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	**R&D Objectives** The project aims to establish a **materials-to-garment engineering framework** that enables blended-yarn design, controlled IR reflectance, hybrid functional finishing, and extreme-environment thermal performance while ensuring durability under realistic military operation conditions. The final goal is to produce **military-ready prototype uniforms and innerwear** that meet domestic defense standards and demonstrate stable performance through field-oriented testing. **Detailed Objectives** ● Development of Functional Fabrics and Prototypes for Combat Uniforms ■ Development of Intrinsically Flame-Retardant Blended Yarns · Develop Aramid/Cotton/Nylon blended yarns with built-in flame-retardant properties. · Optimize blending ratios, yarn counts, and spinning conditions to enable integrated spinning–knitting/weaving–finishing workflows. · Establish fiber/yarn-level FR design to minimize dependence on post-finishing FR treatments. ■ Implementation and Stabilization of Infrared Camouflage Performance · Develop IR-responsive functional dyes and pigments, and identify candidate smart composite materials. · Apply integrated dyeing–printing–finishing packages to control reflectance in the 700–1200 nm IR range. · Design organic–inorganic binders and coatings (e.g., silica-based systems) to ensure long-term IR stability under washing, friction, and UV exposure. ■ Hybrid Organic–Inorganic Finishing for Enhanced FR and Thermal Performance · Apply phosphorus-based FR materials and high-performance additives such as Oxi-PAN, graphene, and CNTs. · Design hybrid inorganic–organic finishing systems incorporating silica and phosphorus-based inorganic components. · Achieve additional FR and thermal performance through synergy between yarn-level intrinsic FR and surface-level finishing. · Enable carbonized-layer formation under heat exposure to suppress flame spread			

and improve wearer protection.

- Structural Fabric Design for Military Uniform Product Categories
  - Develop fabric architectures suited for summer/winter NC combat uniforms and combat shirts.
  - Ensure durability of FR, IR camouflage, and thermal functions after repeated washing, friction, and long-duration wear.
- Development of Base Layer Fabrics for Thermal Innerwear
  - Design lightweight inner-layer fabrics with hygroscopic or photothermal heating capability.
  - Evaluate applicability of low-friction, low-bulk innerwear materials compatible with multi-layer combat uniform systems.
- Prototyping and Performance Verification for General-Purpose Military Use
  - Produce prototypes of summer/winter NC combat uniforms, combat shirts, and thermal innerwear.
  - Validate simultaneous FR + IR camouflage + thermal + mechanical performance under washing, abrasion, and thermal exposure.

- Development of Smart Composite Fabrics and Prototypes for Extreme-Environment Crew Uniforms
  - Development of Environmental-Adaptive Functional Fabrics Using Blended Yarns
    - Develop Aramid/Cotton/Nylon blended yarn–based fabrics optimized for high-temperature, enclosed, and long-duration wear environments.
    - Establish blending, yarn count, and structural design conditions that balance durability and comfort.
    - Ensure compatibility with full spinning–weaving–dyeing–finishing integrated production processes.
  - Development of Combined FR, IR Camouflage, and Thermal Performance
    - Secure fire- and heat-response capability through intrinsically flame-retardant blended yarn structures.
    - Implement IR camouflage through IR-responsive dyes/pigments and controlled reflectance in the 700–1200 nm range.
    - Achieve thermal functions through fabric structure design and surface-treatment strategies.
  - Thermal–Moisture Management for Enclosed, Long-Duration Wear
    - Control heat and moisture transfer to prevent sweat accumulation and thermal entrapment during extended operations in confined environments.
    - Implement microstructural design strategies that provide heating capability while minimizing internal heat buildup.
  - Structural Fabric Design for Special-Purpose Crew Uniforms
    - Design fabric thickness, weight, and flexibility optimized for tracked-vehicle crew operation.

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R&D Target Specifications

- Develop balanced structures that ensure comfort and durability during long-duration wear in enclosed environments.

- Produce design guidelines for fabrics suited to extreme operational conditions.
  - Prototyping and Extreme-Environment Performance Verification
    - Produce prototypes of smart composite crew uniforms.
    - Validate simultaneous FR, IR camouflage, thermal, and mechanical performance under high-temperature, heat-load, washing, and abrasion conditions.
    - Conduct extreme-environment simulation tests to confirm long-term performance stability and reliability.

1. Target Specifications for Functional Combat Uniform Fabrics

Key Performance Indicators		Target	Unit	Evaluation method
Tensile Strength	Warp	≥ 350	N	KS K 0520
	Weft	≥ 350		
Tear Strength	Warp	≥ 20	N	KS K ISO 13937-1
	Weft	≥ 20		
Dimensional Change	Warp	±5	%	KS K ISO 5077, KS K ISO 6330
	Weft	±5		
Flame Resistance	Warp	≤ 3	s	KS K 0585
	Weft	≤ 15	cm	

2. Target Specifications for Tracked-Vehicle Crew Uniform Fabrics

Key Performance Indicators		Target	Unit	Evaluation method
Tensile Strength	Warp	≥ 400	N	KS K 0520
	Weft	≥ 400		
Tear Strength	Warp	≥ 25	N	KS K ISO 13937-1
	Weft	≥ 25		
Dimensional Change	Warp	±3	%	KS K ISO 5077, KS K ISO 6330
	Weft	±3		
Flame Resistance	Warp	≤2	s	KS K 0585
	Weft	≤10	cm	

R&D Plan

The step-by-step research and development plan to achieve the goal is as follows.

Phase 1 (Months 1–6): Securing Core Foundational Textile Technologies

- Develop Aramid/Cotton/Nylon blended yarns with intrinsic flame-retardant performance, and establish optimal spinning conditions (blend ratio, yarn count, twist).
  - Build initial functional fabric samples integrating FR yarns, and validate baseline FR performance, mechanical strength, and wash durability.
  - Identify IR camouflage material candidates (dyes/pigments) and conduct initial reflectance-control finishing tests in the 700–1200 nm IR band.

- Construct early-stage hybrid organic–inorganic finishing formulations (phosphorus-based FR, silica systems) and evaluate adhesion and flexibility.
- Set up an evaluation environment for FR/IR/thermal performance (baseline test protocols, IR reflectance measurement, heat–moisture behavior assessment).

Phase 2 (Months 7–18): Functional Fabric Development and Multi-Property Integration

- Develop integrated IR camouflage finishing (dyeing–printing–coating package) and stabilize IR signatures for military 5-color camouflage patterns.
  - Advance hybrid finishing to simultaneously reinforce flame retardancy and thermal responsiveness; verify performance after repeated washing and abrasion.
  - Establish fabric structural designs for NC combat uniforms (summer/winter) and combat shirts, optimizing strength, air permeability, and comfort.
  - Develop lightweight thermal innerwear fabrics using hygroscopic heating, photothermal absorption, and low-friction base-layer architectures.
  - Produce mid-term prototype fabrics (combat uniform, combat shirt, thermal innerwear) and conduct multi-stress testing (wash/abrasion/heat/UV).
  - Develop first-generation tracked-vehicle crew uniform fabrics and evaluate heat-load, enclosed-environment moisture control, and long-duration comfort.

Phase 3 (Months 19–30): Prototype Production and Mission-Oriented Validation

- Produce full garment prototypes: NC combat uniforms (summer/winter), combat shirts, thermal innerwear, and tracked-vehicle crew uniforms.
  - Conduct comprehensive functional validation: flame retardancy, IR camouflage stability, thermal behavior, mechanical durability, dimensional change.
  - Perform extreme-environment tests for crew uniforms, including high-temperature exposure, sweat–moisture accumulation, and prolonged wear load.
  - Refine hybrid finishing formulations for enhanced long-term stability and optimize fabric structures for comfort and durability in military operations.
  - Finalize design guidelines for each product group (general-purpose uniforms, special-purpose crew uniforms, thermal innerwear).
  - Conduct iterative field-oriented validation with performance tuning based on test results.

Phase 4 (Months 31–36): Reliability Verification and Final Demonstration

- Verify integrated performance and KPI achievement under military-representative test scenarios (FR + IR + thermal + mechanical).
  - Conduct reliability testing after extended wash cycles, friction stress, UV/heat exposure, and multi-layer garment interactions.
  - Finalize prototype garments and prepare documentation for defense procurement evaluation and standard certification.
  - Produce final demonstration samples and compile technical reports, performance data, and commercialization/standardization materials.
  - Confirm production-feasibility and scalability for domestic manufacturing of all

	developed functional fabrics and garments.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Purdue University's Roles and Responsibilities (R&R) ● Conduct foundational research on intrinsically flame-retardant blended yarns, including fiber selection, blend ratio optimization, and thermal degradation analysis. ● Develop and characterize IR camouflage materials (functional dyes/pigments, reflectance-control coatings) and evaluate spectral behavior in the 700–1200 nm range. ● Optimize hybrid organic–inorganic functional finishing systems to reinforce FR, IR, and thermal properties, including compatibility and adhesion testing. ● Conduct extreme-environment simulations (heat load, enclosed-space humidity, long-duration wear load) for tracked-vehicle crew uniform fabrics. ● Establish material-level design principles, performance mechanisms, and reliability criteria required for military-grade functional fabric development. ● Support iterative refinement of prototypes through data-driven performance analysis, structural optimization insights, and material-level troubleshooting.
R&R of Korean SME partners	Korean SME's Role and Responsibilities (R&R) ● Scale-up production of Aramid/Cotton/Nylon blended yarns using optimized spinning, weaving/knitting, and finishing processes. ● Apply IR dyes/pigments, printing patterns, and hybrid finishing treatments using mass-producible dyeing, coating, and curing equipment. ● Develop full fabric structures for NC combat uniforms, combat shirts, thermal innerwear, and tracked-vehicle crew uniforms, integrating design guidelines derived from Purdue's research. ● Produce prototype garments and refine fit, comfort, ergonomics, and multi-layer clothing compatibility suited for military operational use. ● Perform process-level durability testing (wash, abrasion, UV, heat, friction) and incorporate validation outcomes into manufacturable design updates. ● Lead pilot production, pre-commercial manufacturing trials, and domestic supply-chain coordination for future mass production.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	● Technical Outcomes: Establishing an Integrated Functional Military Textile Platform and Performance Verification Basis ■ Development of a complete materials-to-garment engineering platform that integrates intrinsic flame-retardant blended yarns, IR camouflage treatments (700–1200 nm), hybrid organic–inorganic finishing, and adaptive thermal fabrics, enabling stable multi-functional performance suitable for military environments. ■ Establishment of testing protocols for IR reflectance evaluation, flame-retardant

	durability, thermal–moisture transport, and extreme-environment wear, supported by joint verification using Purdue's advanced textile testing infrastructure. ■ Production of fully validated prototype garments—including NC combat uniforms (summer/winter), combat shirts, thermal innerwear, and tracked-vehicle crew uniforms—and secure complete performance reports aligned with defense procurement standards. ■ Generation of core intellectual property related to intrinsic FR yarn design, IR camouflage coating formulations, hybrid finishing chemistry, and military textile structural architectures, supporting domestic and future international protection. ● Economic and industrial impact ■ Strengthening of Korean SME's manufacturing capability by securing core FR/IR/Thermal textile technologies previously sourced from overseas suppliers. ■ Expansion of Korean SMEs' production capacity into high-value defense textiles, industrial safety clothing, firefighting gear, and extreme-environment workwear. ■ Enhancement of export competitiveness through development of globally credible functional textile technologies with potential applicability to allied defense markets. ■ Formation of a scalable textile technology stack enabling efficient pilot production, supply-chain integration, and transition to mass manufacturing. ■ Realization of import substitution effects by replacing foreign FR and IR fabrics with domestically developed textiles. ● Social and environmental impact ■ Improvement of soldier safety through enhanced flame protection, reduced IR detectability, and better thermal comfort in extreme environments. ■ Support for mission endurance and operational effectiveness through optimized heat–moisture management and long-duration wear comfort. ■ Reinforcement of national defense resilience via self-reliant protective textile technologies and stable domestic supply capacity. ■ Extension of dual-use benefits to firefighters, industrial workers, emergency responders, and outdoor professionals through safer and more advanced protective garments. ■ Reduction of environmental burden through durable, long-life functional fabrics that minimize waste and replacement frequency.

6. References